



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO Sapthagiri Balaji Agency

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

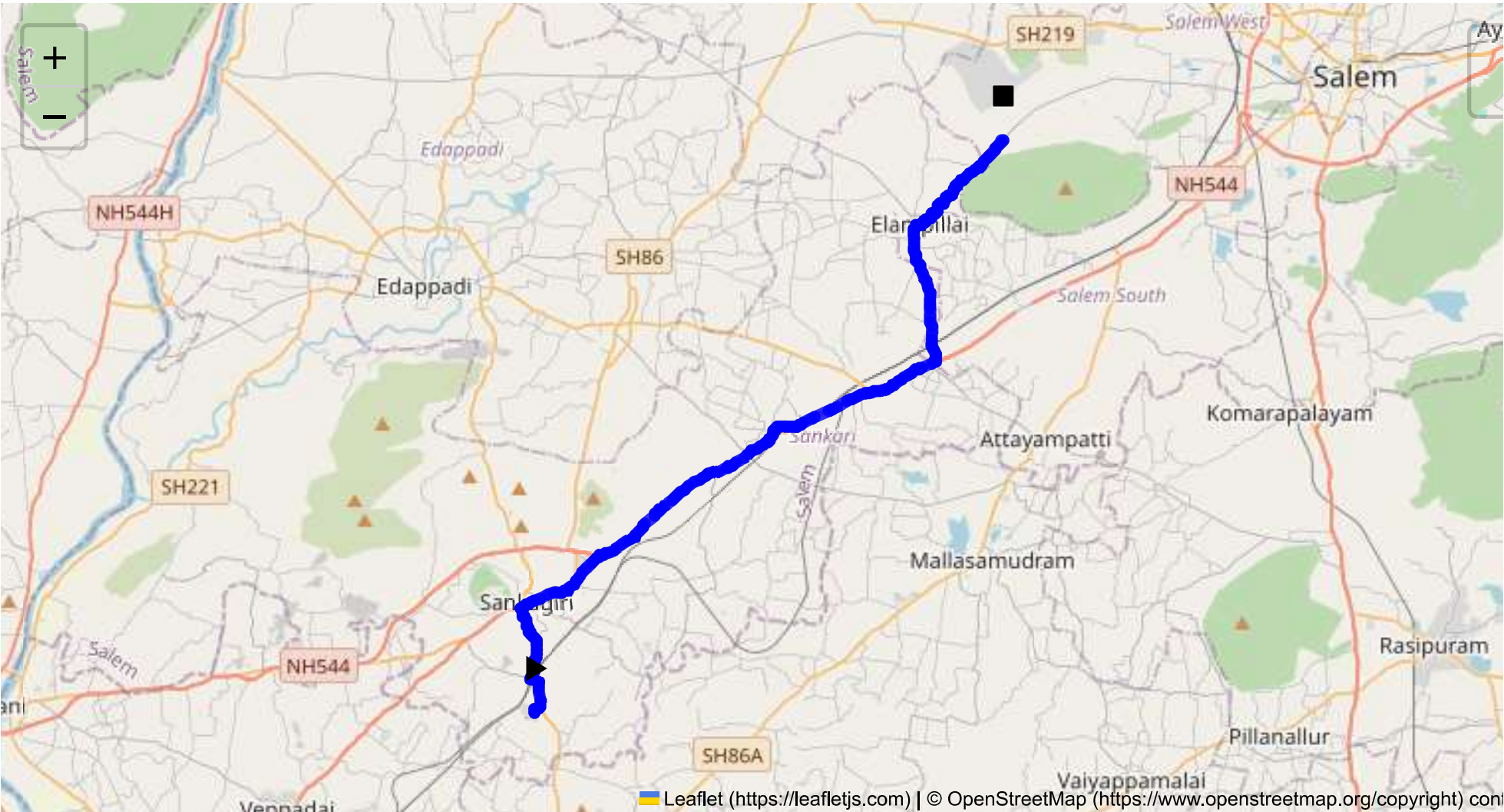
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 33.51 km
Estimated Duration: 0.8 hours
Adjusted Duration (Heavy Vehicle): 1.1 hours
Start: (11.4381, 77.8734)
End: (11.633233, 78.035332)



Welcome to the Journey Risk Management Study

Route Analysis from CVQF+23W, Sangagiri to 1, Siddhar Kovil Rd, Salem

1. Overview of the Route Map

The route from Sangagiri to Salem covers approximately 33.51 kilometers and is primarily along the NH 544 highway. It is a straightforward route with a few bends and small junctions. The NH 544 is a well-maintained national highway, considered a vital connective corridor in Tamil Nadu.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. Along this route:

- **Summer (March to June):** High temperatures, risk of road surface melting.
- **Monsoon (July to September):** Heavy rains can cause temporary flooding and reduced visibility.
- **Winter (October to February):** Mild conditions with occasional fog early in the morning. Drivers should be prepared for reduced visibility and slippery roads during the monsoon season.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically 8:00 AM - 10:00 AM and 5:00 PM - 8:00 PM due to local commuter traffic.
- **Congestion-Prone Areas:** Key congestion zones include the main junctions near Salem where local traffic increases.
- **Holiday and Festival Delays:** Expect increased traffic during regional festivals.

4. Assessment of Road Quality and Infrastructure

The NH 544 is generally in good condition with frequent signage and adequate lane width for heavy vehicles.

- **Potential Hazards:** Potholes and uneven road surfaces may occur, especially post-monsoon.

5. Suggestions for Alternative Routes

In case of emergencies or road closures, consider using state highways like the SH 86, which provides a parallel route, albeit slightly longer.

6. Local Regulations Affecting Hazardous Material Transport

- **Permits:** Ensure all necessary permits are in place for transporting hazardous materials.
- **Vehicle Restrictions:** Some local roads in Salem may have restrictions on heavy vehicle movement during peak hours.
- **Safety Checks:** Mandatory safety checks and documentation.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Reliable historical data mentions occasional incidents related to overturn accidents, largely attributed to speed or sharp turns. No major hazardous material spillage incidents recorded recently.

8. Environmental Considerations and Sensitive Areas

- **Protected Zones:** No immediate wildlife sanctuaries or protected zones within this route.
- **Pollution Control:** Ensure minimal emissions as the road approaches urban settlements, particularly near Salem.

9. Analysis of Communication Coverage

Mobile network coverage along NH 544 is generally good due to proximity to urban areas. Possible dead zones are typically near underpasses but are very short.

10. Estimated Emergency Response Times

Given the route's proximity to Salem:

- **Rural Segments:** Approximately 30-45 minutes response time due to less dense hospital and enforcement presence.
- **Urban Proximity:** 15-30 minutes, as Salem has several emergency facilities.

11. Overall Summary of Risk Assessment

The route from Sangagiri to Salem is largely safe for heavy vehicle transport provided there is adherence to regulations and seasonal weather advisories. The NH 544 is well-maintained, but peak traffic periods and monsoon conditions require heightened caution. Alternative routes are available for circumstantial detours, and emergency response times are reasonable. Regular updates on traffic conditions and weather alerts are essential for minimizing risks related to the transport of hazardous materials on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	Medium	11.48366, 77.88781	30 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.55839, 78.01288	15 KM/Hr
10	Turn	Medium	11.60320, 78.00504	30 KM/Hr
11	Turn	Medium	11.60479, 78.00610	30 KM/Hr
12	Turn	High	11.60552, 78.00827	15 KM/Hr
13	Turn	Medium	11.60547, 78.00856	30 KM/Hr
14	Turn	Medium	11.60559, 78.00883	30 KM/Hr
15	Turn	Medium	11.60747, 78.01075	30 KM/Hr
16	Turn	High	11.61170, 78.01514	15 KM/Hr
17	Turn	Medium	11.61176, 78.01527	30 KM/Hr
18	Turn	Medium	11.61440, 78.01791	30 KM/Hr
19	Blind Spot	Blind Spot	11.63330, 78.03570	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Vembadithalam Government General Hospital	11.5683, 78.01211	30 km/h	Medium